

2.6. Electromagnets, induction & inductance

em17 - Thomson

Where does the energy come from?

New concepts:

- magnetic flux & induction
- electromagnets & transformers
- alternating current (AC)



Interaction of coil & magnet

- previous lecture we used **constant** magnetic field to understand:
 1. electric current produces magnetic field
 2. magnetic field exerts a force on electric currents/moving electric particles

What happens when we change the magnetic field over time?

em18 - magnet drop

- Joseph Henry (1797-1878) & Michael Faraday (1791-1867) described independently that **changing magnetic field induces an electric field**
- **concepts to understand:**
 - alternating voltage/current
 - electromagnetic induction
 - inductance & electromagnets

What influences the induced voltage ? (1/2)

emf - flux

Impact of geometry and orientation

The flux strikes back: Magnetic flux

sim mag flux

- magnetic flux Φ_B defined as:

$$\Phi_B = \int \vec{\mathbf{B}} d\vec{\mathbf{A}}$$

- here $\vec{\mathbf{A}}$ is the vector normal to the area A
- in contrast to the electric flux Φ_E , "regular" integral instead of closed surface integral
- for a **uniform field** simplifies to:

$$\Phi_B = B_{\perp} A = BA \cos \theta = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}}$$

- measured in *weber* [Wb] = [T m²]

What influences the induced voltage ? (2/2)

em12 - flux

Impact of change over time and number of loops

- **only changing magnetic fields induce a voltage**
- rate of change scales with induced voltage, i.e. $\frac{d\Phi_B}{dt}$
- for same $\frac{d\Phi_B}{dt}$, coil with more loops shows higher induced voltage

Faraday's law of induction

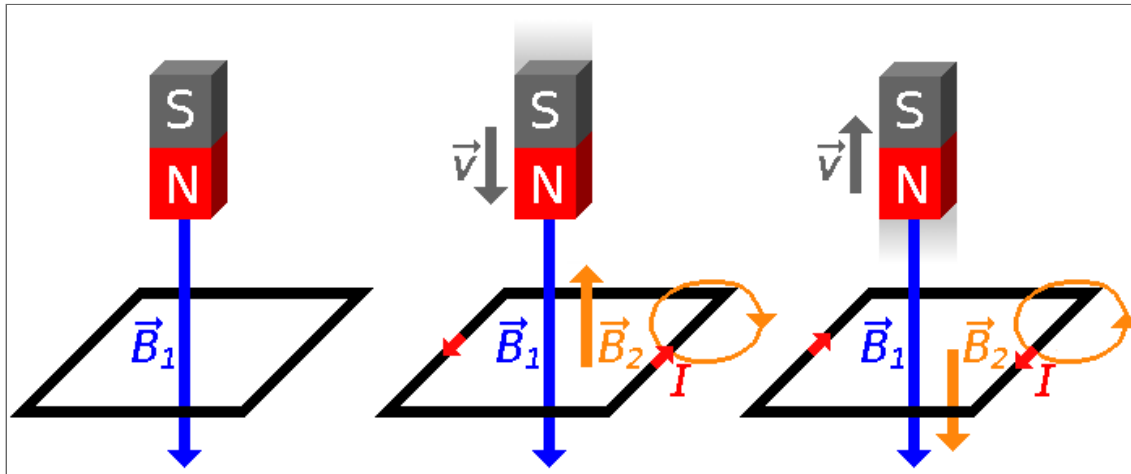
- $V_{ind} = -N \frac{d\Phi_B}{dt}$
- with:
 - N - number of loops in coil
 - $\frac{d\Phi_B}{dt}$ - change in magnetic flux

Why is there a minus?

Lenz's law

em14 - Rings

A change in the magnetic flux through a conducting loop induces a voltage, so that the resulting current generates a magnetic field that opposes the change in magnetic flux.



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E-field from electrostatics vs. electromagnetism

Electrostatics	Electromagnetism
static charges generate electric fields	changing magnetic flux induces electric fields
field lines start/end on charges	field lines can form closed loops
voltage between two points: $V_{AB} = - \int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}}$	induced voltage: $V_{ind} = - \frac{d\Phi_B}{dt}$
$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = 0$	$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = - \frac{d\Phi_B}{dt}$ (Faradys law)
conservative field / path independent	non-conservative field / closed-loop work can be non-zero
• electrostatics: ■ after returning to the starting point, a charge has no net energy gain ■ positive and negative work cancel exactly	

- changing magnetic fields:
 - induced electric field continuously pushes charges around the loop
 - force on the charge always points along the loop
 - after one full loop, charges can gain energy

Eddy currents

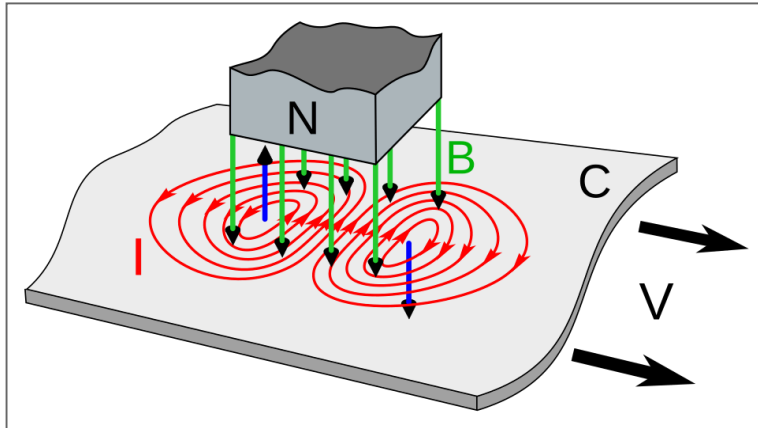
em25 - Waltenhofen's pendulum

- conductor (pendulum) moves through magnetic field generated by an electromagnet
- → changing magnetic flux in conductor → induced eddy currents
- part leaving the magnetic field:
 - decrease in magnetic flux (number of field lines passing through)
 - induced currents generate magnetic fields opposing the decrease (Lenz's law)
 - resulting force pulls the conductor back toward the electromagnet
- part entering the magnetic field:
 - increase in magnetic flux (number of field lines passing through)
 - induced currents generate magnetic fields opposing the increase (Lenz's law)

- resulting force pushes the conductor away from the electromagnet

Eddy currents (cont')

em25 - Waltenhofen's pendulum



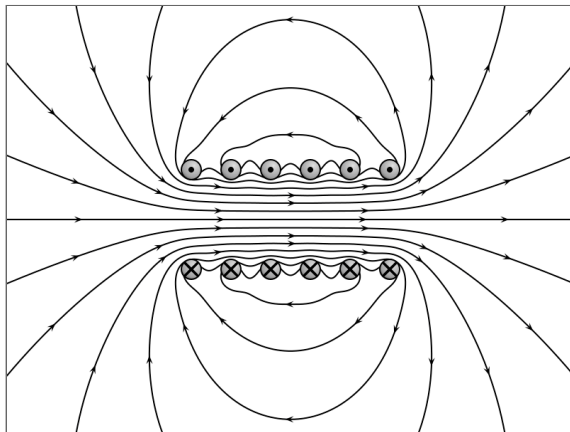
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- direction of B via RHR-1 by matching curl of fingers to induced current

Electromagnets - solenoids

em10 + simulation B-field solenoid

- tightly wound helical coils of wire
- found in many devices: generators, loudspeakers, MRI scanners, and many more
- produce an almost uniform magnetic field inside the coil and weak field outside
- with loop density n , the field is: $B = \mu_0 \frac{N}{l} I = \mu_0 n I$



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From induction to alternating current

- Faraday's law: **changing magnetic flux induces voltage**

$$V_{ind} = -\frac{d\Phi_B}{dt}$$

- $\rightarrow$ if magnetic flux changes periodically, **induced voltage also changes periodically**
- induced voltage therefore naturally alternates between positive and negative values

$$V(t) = V_0 \sin(\omega t)$$

- this is called **alternating current (AC)**

Primer on alternating current (AC)

- AC = periodically changing voltage and current
- sinusoidal AC is the most important case:

$$V(t) = V_0 \sin(2\pi ft) = V_0 \sin(\omega t)$$

$$I(t) = I_0 \sin(\omega t)$$

- angular frequency:

$$\omega = 2\pi f$$

- standard power grid:
 - most of the world: 50 Hz
 - north America: 60 Hz

Using Root-Mean-Squared (RMS) values

- often RMS values are used in AC:

$$V_{\text{RMS}} = \sqrt{V(t)^2} = \sqrt{V_0^2 \sin^2(\omega t)}$$

- with $\sin^2(\omega t) = \frac{1}{2}$

$$V_{\text{RMS}} = \frac{V_0}{\sqrt{2}}$$

$$I_{\text{RMS}} = \frac{I_0}{\sqrt{2}}$$

- for example, a 220V outlet has a peak voltage $V_0 \approx 311 \text{ V}$

Power in AC circuits

- power is $P(t) = I(t)V(t)$
- for sinusoidal current/voltage, average power over a period is **non-zero**:

$$\bar{P} = I_0 \sin(\omega t) V_0 \sin(\omega t) = I_0 V_0 \sin^2(\omega t)$$

- for one period $\sin^2(\omega t) = \frac{1}{2}$, $I_{RMS} = \frac{I_0}{\sqrt{2}}$ and $V_{RMS} = \frac{V_0}{\sqrt{2}}$

$$\bar{P} = \sqrt{2}I_{RMS}\sqrt{2}V_{RMS}\frac{1}{2} = I_{RMS}V_{RMS}$$

Transformers

em46

- transformer consists of a *primary* and *secondary* coil
- two coils **do not** form a closed circuit, but only their proximity is essential for the transformer to work

What influences the voltage in the secondary coil?

Transformers - What influences the voltage in the secondary coil?

- iron core (geometry, layered, etc.)
- ratio of loops:
 - without energy loss, the induced voltages in both coils are:

$$V_2 = N_2 \frac{d\Phi_B}{dt} \quad \& \quad V_1 = N_1 \frac{d\Phi_B}{dt}$$

- taking the voltage ratio & using conservation of power, $P_1 = P_2$, we get:

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

- depending on the ratio of the loop numbers:
 - step-up transformer: For $N_2 > N_1$ the voltage V_2 will be larger than V_1

- step-down transformer: For $N_2 < N_1$ the voltage V_2 will be smaller than V_1

Mutual inductance

- relates the induced voltage in one coil to the rate of change of current in a neighboring coil
- defined as proportionality constant:

$$M_{21} = \frac{N_2 \Phi_{21}}{I_1}$$

- the induced voltage in one coil can directly be related to change in current in the other coil:

$$V_2 = -N_2 \frac{d\Phi_{21}}{dt} = -M_{21} \frac{dI_1}{dt} \quad \& \quad V_1 = -M_{12} \frac{dI_2}{dt}$$

- mutual inductance only depends on geometric variables, i.e. the spacing between the coils and their individual shape, size, and number of loops:

$$M = M_{21} = M_{12}$$

- **inductance** is expressed in **units of henry** $[H] = [V \text{ s} / A] = [\Omega \text{ s}]$

Self-inductance & inductors

YouTube - James Bond

em13

- even for a **single coil**, we find the phenomenon of inductance, a.k.a **self-inductance**
- **changing current** passes through the coil **induces voltage in coil itself** with **opposed direction**
- defined as proportionality constant (depends on coil geometry):

$$L = \frac{N\Phi_B}{I}$$

$$V_{ind} = -N \frac{d\Phi_B}{dt} = -L \frac{dI}{dt}$$

- often referred simply to as inductance and measured in henry [H].

- **inductors are fundamental circuit elements alongside resistors and capacitors**

Revisit initial experiment & energy stored in the magnetic field

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- power supplied to an inductor is:

$$P = I V_{\text{ind}} = I L \frac{dI}{dt}$$

- the work done to build up the current is:

$$dW = P dt = I L \frac{dI}{dt} dt = I L dI$$

$$W = \int_0^I I L dI = \frac{1}{2} L I^2$$

- energy stored in the magnetic field is: $U = \frac{1}{2} L I^2$

- analogous to the energy stored in a capacitor: $U = \frac{1}{2}CV^2$